Reasoning test

For admission to Class 11

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 10 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 10.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, on these topics.

Maths

1. Number patterns and series
2. Primes, factors, HCF and LCM
3. Fractions
4. Perimeter and area
5. Simple equations
6. Speed, distance and time
7. Averages
8. Ratio and proportion
9. Percentages, profit and loss, interest
10. Squares and cubes
11. Probability

English

1. Vocabulary
2. Grammar
3. Sentence completion and ordering
4. Reading comprehension

Science

1. Evaporation and condensation
2. Magnets
3. Light, shadows and mirrors
4. Acids, bases and indicators
5. Physical and chemical changes
6. Heat
7. Electric circuits
8. Respiration
9. Photosynthesis
10. Motion and speed
11. Food chains
12. Floating, sinking and density
13. Force and pressure
14. Sound
15. Mass and weight
16. Force and laws of motion
17. Change of state
18. Elements, compounds and mixtures
19. Energy
20. Electric power and safety

Reasoning

1. Clocks and calendars
2. Number codes and puzzles
3. Everyday units
4. Word problems
5. Time and work
6. Problems on ages
7. Mixtures
8. Venn diagrams
9. Analogies, odd one out, series and codes
10. Logical deduction
11. Reading an experiment
12. Applying a principle to a new situation

Class 10 syllabus questions

Maths

1. Real Numbers
2. Polynomials
3. Pair of Linear Equations in Two Variables
4. Quadratic Equations
5. Arithmetic Progressions
6. Triangles
7. Coordinate Geometry
8. Introduction to Trigonometry

Science

1. Chemical Reactions and Equations
2. Acids, Bases and Salts
3. Life Processes
4. Control and Coordination
5. Light – Reflection and Refraction
6. Electricity

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   Which of these numbers is divisible by both 3 and 4?

   A 132 **B** 138
   C 142 **D** 150

2. ENGLISH · BASICS
   The sentries stayed vigilant all night. As used here, "vigilant" most nearly means

   A tired **B** watchful
   C silent **D** brave

3. SCIENCE · BASICS
   A 1,000 W room heater is switched on for 3 hours. How many units (kilowatt-hours) of electricity does it use?

   Type-in answer: _____

4. MATHS · CLASS 10
   What is the HCF of 24 and 36?

   A 6 **B** 72
   C 4 **D** 12

5. ENGLISH · BASICS
   Choose the correct word: The cadets were proud ___ their achievement.

   A of **B** for
   C with **D** at

6. SCIENCE · CLASS 10
   Which gas is given off when zinc reacts with dilute hydrochloric acid?

   A Oxygen **B** Hydrogen
   C Carbon dioxide **D** Chlorine

7. MATHS · BASICS
   Kiran scored 54 marks out of 75 in a test. What percentage did Kiran score?

   Type-in answer: _____

8. ENGLISH · BASICS
   Which word is the odd one out?

   A Cricket **B** Hockey
   C Football **D** Chess

9. SCIENCE · BASICS
   A fuse is fitted in a household circuit so that, if too much current flows, it

   A raises the voltage to appliances **B** melts and breaks the circuit
   C stores the extra electricity **D** makes the bulbs glow brighter

10. MATHS · CLASS 10
    What is the 15th term of the arithmetic progression 5, 8, 11, …?

    Type-in answer: _____

11. ENGLISH · BASICS
Choose the word most nearly opposite in meaning to "humble".

A proud
B kind
C quiet
D poor

12. SCIENCE · CLASS 10
Which blood vessels carry blood away from the heart to the rest of the body?

A Arteries
B Veins
C Capillaries
D Nerves

13. MATHS · BASICS
A bus leaves at 9:40 am and travels 150 km at a steady 60 km/h. At what time does it arrive?

A 11:40 am
B 11:50 am
C 12:10 pm
D 12:30 pm

14. ENGLISH · BASICS
Choose the correct words: She ___ in this school since 2022.

A studies
B is studying
C has studied
D studied

15. SCIENCE · BASICS
Which of these is a compound?

A water
B oxygen
C iron
D air

16. MATHS · CLASS 10
What is the value of tan 45°?

A 0
B 1
C 1/√2
D √3

17. ENGLISH · BASICS
To "burn the midnight oil" means to

A spend money carelessly
B work late into the night
C light a lamp at midnight
D forget an important task

18. SCIENCE · CLASS 10
A 10 Ω resistor is connected across a 5 V battery. What current flows through it, in amperes?

Type-in answer: _____

19. MATHS · BASICS
After a 25% increase, a salary is ₹20,000 a month. What was it before the increase?

A ₹15,000
B ₹16,000
C ₹16,500
D ₹17,500

20. ENGLISH · BASICS
If COLD is written as DPME in a code, how is WARM written in the same code?

A VZQL
B XBSM
C XBSN
D YCTO

21. SCIENCE · BASICS
A 60 W bulb and a 1,500 W heater are each used for 2 hours. How many times more energy does the heater use?

A 2.5 times
B 12.5 times
C 25 times
D 50 times

22. MATHS · CLASS 10

What are the roots of the equation $x^2 - 5x + 6 = 0$?

A 2 and 3
B −2 and −3
C 1 and 6
D −1 and 6

23. ENGLISH · BASICS

"Carry a water bottle on the march; there are no taps on the route." This advice assumes that

A the marchers will need water
B taps on routes are often dirty
C the march will be a short one
D water bottles are not costly

24. SCIENCE · CLASS 10

A ray of light travels at an angle from air into glass. At the surface, the ray bends

A away from the normal
B along the surface
C towards the normal
D back into the air

25. MATHS · BASICS

Two fair dice are thrown together. What is the probability that the two numbers add up to 8?

A 1/6
B 1/9
C 7/36
D 5/36

26. ENGLISH · BASICS

All cadets wake at 5 am. Nobody who wakes at 5 am misses breakfast. Arun misses breakfast. Which statement must be true?

A Arun wakes at 5 am
B Arun is a cadet
C Some cadets miss breakfast
D Arun is not a cadet

27. SCIENCE · BASICS

A wooden block floats in water with exactly half of its volume under the surface. Water has a density of 1 g/cm³. What is the density of the wood?

A 1 g/cm³
B 2 g/cm³
C 0.25 g/cm³
D 0.5 g/cm³

28. MATHS · CLASS 10

The point (5, 4) lies exactly halfway between (2, a) and (b, 6). Find a + b.

Type-in answer: _____

29. ENGLISH · BASICS

P is taller than Q. R is shorter than Q. S is taller than P. Who is the second tallest?

A S
B Q
C P
D R

30. SCIENCE · CLASS 10

Three 6 Ω resistors are connected in parallel across a 12 V battery. What total current is drawn, in amperes?

Type-in answer: _____

Answers

1	**A. 132**	132 = 12 × 11, so it is divisible by both. 138 and 150 are divisible by 3 but not by 4; 142 is divisible by neither.
2	**B. watchful**	Vigilant means keeping careful watch for danger, which is exactly what a sentry does. Tired or silent sentries may still be vigilant.
3	**3**	1,000 W is 1 kW, and energy is power × time: 1 kW × 3 h = 3 kWh, which is 3 units. This is a type-in question.
4	**D. 12**	$24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$. The HCF takes the lowest power of each common prime: $2^2 \times 3 = 12$. 72 is the LCM.
5	**A. of**	"Proud" is followed by "of": proud of something. The other prepositions are not used with proud in this sense.
6	**B. Hydrogen**	A metal and a dilute acid give a salt and hydrogen: zinc chloride and hydrogen here. Carbon dioxide comes from acids with carbonates.
7	**72**	54 ÷ 75 = 0.72, which is 72%. Another way: 75 marks is 100%, so each mark is 4/3 %, and 54 × 4/3 = 72. This is a type-in question.
8	**D. Chess**	Cricket, hockey and football are outdoor team games played with a ball. Chess is a board game for two players.
9	**B. melts and breaks the circuit**	A fuse wire melts when the current is too large, which breaks the circuit and stops the flow before wires overheat or catch fire.
10	**47**	a = 5 and d = 3, so the 15th term is a + 14d = 5 + 42 = 47. This is a type-in question: the answer is typed as a number.
11	**A. proud**	Humble means modest, not thinking oneself important, so the opposite is proud. A humble person may also be kind or quiet, but those are not opposites.
12	**A. Arteries**	Arteries carry blood away from the heart under high pressure; veins bring it back. Capillaries link the two, and nerves carry signals, not blood.
13	**C. 12:10 pm**	The journey takes 150 ÷ 60 = 2.5 hours, which is 2 hours 30 minutes. 9:40 am plus 2 hours 30 minutes is 12:10 pm.
14	**C. has studied**	"Since 2022" marks an action that began in the past and continues now, which takes the present perfect: "has studied". The simple present and past do not fit "since".
15	**A. water**	Water is hydrogen and oxygen chemically joined, so it is a compound. Oxygen and iron are elements, and air is a mixture.
16	**B. 1**	In a right triangle with a 45° angle the two shorter sides are equal, so tan 45° = opposite ÷ adjacent = 1. $1/\sqrt{2}$ is sin 45° or cos 45°.
17	**B. work late into the night**	The idiom comes from working by lamplight after midnight: it means staying up late to work or study. Lighting the lamp is the picture, not the meaning.
18	**0.5**	By Ohm's law I = V ÷ R = 5 V ÷ 10 Ω = 0.5 A. This is a type-in question: the answer is typed as a number.

19	**B. ₹16,000**	The new salary is 125% of the old one, so the old salary is 20,000 ÷ 1.25 = ₹16,000. Taking 25% off ₹20,000 gives the wrong ₹15,000.
20	**C. XBSN**	Each letter moves one step forward: C→D, O→P, L→M, D→E. So W→X, A→B, R→S, M→N, giving XBSN.
21	**C. 25 times**	Both run for the same time, so energy goes up with power alone: 1,500 ÷ 60 = 25. The 2 hours cancels out.
22	**A. 2 and 3**	$x^2 - 5x + 6 = (x - 2)(x - 3)$, so the roots are 2 and 3. 1 and 6 multiply to 6 but add to 7, not 5.
23	**A. the marchers will need water**	The advice only makes sense if the marchers will need water along the way. The other statements are not needed for it to hold.
24	**C. towards the normal**	Light slows down in glass, a denser medium, so it bends towards the normal. Going from glass into air it bends away from the normal.
25	**D. 5/36**	There are 36 equally likely outcomes. A total of 8 comes from (2,6), (3,5), (4,4), (5,3) and (6,2), which is 5 outcomes: 5/36.
26	**D. Arun is not a cadet**	Every cadet wakes at 5 am, and everyone who does has breakfast. Arun does not, so Arun cannot wake at 5 am and so cannot be a cadet.
27	**D. 0.5 g/cm³**	A floating block pushes aside its own weight of water. It does so with half its volume, so it weighs as much as half its volume of water: 0.5 g/cm³.
28	**10**	Midpoint x: (2 + b)/2 = 5, so b = 8. Midpoint y: (a + 6)/2 = 4, so a = 2. Then a + b = 10. This is a type-in question.
29	**C. P**	Putting the statements together gives S > P > Q > R, so P is second tallest. S is the tallest and R the shortest.
30	**6**	In parallel 1/R = 3 × 1/6 = 1/2, so R = 2 Ω and I = 12 V ÷ 2 Ω = 6 A. Treating them in series (18 Ω) gives 0.67 A.